PERMETHRIN

Scabies first-line
Treat all contacts

CHEMICAL NAME 3-phenoxybenzyl (1RS,3RS;1RS,3SR)-3-(2,2-dichlorovinyl)-2,2-dimethylcyclopropanecarboxylate
CLASS Pyrethroid ectoparasiticide (insecticide)
TRADE NAMES Lyclear · Elimite · Nix (head lice) · Permethrin 5% cream

THERAPEUTIC USES / INDICATIONS

- Scabies (first-line, including pregnancy and children ≥2 months).
- Head lice (1% rinse).
- Pubic lice.

ADMINISTRATION

- **Scabies:** apply 5% cream to whole body from neck down (include scalp, face, neck, ears in elderly, infants, immunocompromised), especially between fingers/toes, under nails, genitals; wash off after 8–12 h; repeat after 7 days.
- Reapply to hands if washed during treatment.
- Treat all household and close contacts simultaneously.
- Wash bedding, clothes, towels at ≥60 °C.

DOSAGES

- **Scabies (adult):** 30 g of 5% cream (one tube) per application; repeat after 7 days.
- **Children:** 2 months–1 yr ~3.75 g; 1–5 yr 7.5 g; 6–12 yr 15 g.
- **Head lice:** 1% cream rinse on washed, towel-dried hair for 10 min; repeat in 7 days.

PHARMACOKINETICS

- <2% absorbed through skin; rapidly hydrolysed to inactive metabolites; excreted in urine.

PHARMACODYNAMICS

- Disrupts **sodium channels** in parasite nerve membranes → delayed repolarisation → paralysis and death of mites/lice and eggs.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Itching and burning (pruritus may persist 2–4 weeks after successful treatment).
- Erythema, tingling.
- Rash.

ADVERSE EFFECTS (SERIOUS)

- Contact dermatitis.
- Treatment failure (resistance, incorrect application).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to pyrethroids or chrysanthemums.

KEY INTERACTIONS

- None significant.

NURSING MANAGEMENT

- Teach correct application; supervise in institutions.
- Coordinate treatment of all contacts and environmental cleaning.
- Infection control (contact precautions in hospital/care homes).
- Consider crusted scabies (needs oral ivermectin + topical).

PATIENT EDUCATION

- Apply to cool, dry skin (not after a hot bath) all over the body; leave 8–12 h, then wash off.
- Repeat after 1 week; treat everyone in the household at the same time.
- Itch may continue for up to 4 weeks — this does not mean treatment failed.
- Wash bedding/clothing hot or seal in bags for 72 h.

KEY POINTS

- Pyrethroid; scabies and lice.
- Neck down, wash off after 8–12 h, repeat at 7 days.
- Treat all contacts; launder at 60 °C.
- Itch persists up to 4 weeks.